

Quantium

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R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
library(tidyverse)
```

```
## Warning: package 'tidyr' was built under R version 4.5.3
```

```
## Warning: package 'dplyr' was built under R version 4.5.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.1      v readr      2.1.6
```

```
## v forcats    1.0.1      v stringr    1.6.0
```

```
## v ggplot2    4.0.2      v tibble     3.3.1
```

```
## v lubridate  1.9.5      v tidyr      1.3.2
```

```
## v purrr      1.2.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(data.table)
```

```
## Warning: package 'data.table' was built under R version 4.5.3
```

```
##
```

```
## Attaching package: 'data.table'
```

```
##
```

```
## The following objects are masked from 'package:lubridate':
```

```
##
```

```
##      hour, isoweek, isoyear, mday, minute, month, quarter, second, wday,
```

```
##      week, yday, year
```

```
##
```

```
## The following objects are masked from 'package:dplyr':
```

```
##
```

```
##      between, first, last
```

```
##
## The following object is masked from 'package:purrr':
##
## transpose
```

```
library(ggplot2)
library(readr)
library(dplyr)
filePath <- "F:/QVI/"
transactionData <- fread(paste0(filePath,"QVI_transaction_data.csv"))
customerData <- fread(paste0(filePath,"QVI_purchase_behaviour.csv"))
str(transactionData)
```

```
## Classes 'data.table' and 'data.frame': 264836 obs. of 8 variables:
## $ DATE : int 43390 43599 43605 43329 43330 43604 43601 43601 43332 43330 ...
## $ STORE_NBR : int 1 1 1 2 2 4 4 4 5 7 ...
## $ LYLTY_CARD_NBR: int 1000 1307 1343 2373 2426 4074 4149 4196 5026 7150 ...
## $ TXN_ID : int 1 348 383 974 1038 2982 3333 3539 4525 6900 ...
## $ PROD_NBR : int 5 66 61 69 108 57 16 24 42 52 ...
## $ PROD_NAME : chr "Natural Chip Compny SeaSalt175g" "CCs Nacho Cheese 175g" "Smiths ...
## $ PROD_QTY : int 2 3 2 5 3 1 1 1 1 2 ...
## $ TOT_SALES : num 6 6.3 2.9 15 13.8 5.1 5.7 3.6 3.9 7.2 ...
## - attr(*, ".internal.selfref")=<externalptr>
```

```
str(customerData)
```

```
## Classes 'data.table' and 'data.frame': 72637 obs. of 3 variables:
## $ LYLTY_CARD_NBR : int 1000 1002 1003 1004 1005 1007 1009 1010 1011 1012 ...
## $ LIFESTAGE : chr "YOUNG SINGLES/COUPLES" "YOUNG SINGLES/COUPLES" "YOUNG FAMILIES" "OLDER SI ...
## $ PREMIUM_CUSTOMER: chr "Premium" "Mainstream" "Budget" "Mainstream" ...
## - attr(*, ".internal.selfref")=<externalptr>
```

```
transactionData$DATE <- as.Date(transactionData$DATE, origin = "1899-12-30")
summary(as.factor(transactionData$PROD_NAME))
```

```
## Kettle Mozzarella Basil & Pesto 175g
## 3304
## Kettle Tortilla ChpsHny&Jlpno Chili 150g
## 3296
## Cobs Popd Swt/Chlli &Sr/Cream Chips 110g
## 3269
## Tyrrells Crisps Ched & Chives 165g
## 3268
## Cobs Popd Sea Salt Chips 110g
## 3265
## Kettle 135g Swt Pot Sea Salt
## 3257
## Tostitos Splash Of Lime 175g
## 3252
## Infuzions Thai SweetChili PotatoMix 110g
## 3242
## Smiths Crnkle Chip Orgnl Big Bag 380g
```

##		3233
##	Thins Potato Chips Hot & Spicy	175g
##		3229
##	Kettle Sensations Camembert & Fig	150g
##		3219
##	Doritos Corn Chips Cheese Supreme	170g
##		3217
##	Pringles Barbeque	134g
##		3210
##	Doritos Corn Chip Mexican Jalapeno	150g
##		3204
##	Kettle Sweet Chilli And Sour Cream	175g
##		3200
##	Smiths Crinkle Chips Salt & Vinegar	330g
##		3197
##	Thins Chips Light& Tangy	175g
##		3188
##	Dorito Corn Chp Supreme	380g
##		3185
##	Pringles Sweet&Spcy BBQ	134g
##		3177
##	Infuzions BBQ Rib Prawn Crackers	110g
##		3174
##	Tyrrells Crisps Lightly Salted	165g
##		3174
##	Kettle Sea Salt And Vinegar	175g
##		3173
##	Doritos Corn Chip Southern Chicken	150g
##		3172
##	Twisties Chicken	270g
##		3170
##	Twisties Cheese Burger	250g
##		3169
##	Grain Waves Sweet Chilli	210g
##		3167
##	Pringles SourCream Onion	134g
##		3162
##	Doritos Corn Chips Nacho Cheese	170g
##		3160
##	Cobs Popd Sour Crm &Chives Chips	110g
##		3159
##	Kettle Original	175g
##		3159
##	Pringles Original Crisps	134g
##		3157
##	Cheezels Cheese	330g
##		3149
##	Kettle Honey Soy Chicken	175g
##		3148
##	Kettle Tortilla ChpsBtroot&Ricotta	150g
##		3146
##	Tostitos Smoked Chipotle	175g
##		3145
##	Infzns Crn Crnchers Tangy Gcamole	110g

##			3144
##	Smiths Crinkle	Original	330g
##			3142
##	Kettle Tortilla Chps	Feta&Garlic	150g
##			3138
##	Infuzions SourCream&Herbs	Veg Strws	110g
##			3134
##	Kettle Sensations	Siracha Lime	150g
##			3127
##	Old El Paso Salsa	Dip Chnky Tom Ht	300g
##			3125
##	Doritos Corn Chips	Original	170g
##			3121
##		Doritos Mexicana	170g
##			3115
##		Twisties Cheese	270g
##			3115
##	Old El Paso Salsa	Dip Tomato Med	300g
##			3114
##	Pringles Mystery	Flavour	134g
##			3114
##	Thins Chips Seasoned	chicken	175g
##			3114
##	Grain Waves Sour	Cream&Chives	210G
##			3105
##	Pringles Chicken	Salt Crips	134g
##			3104
##	Thins Chips Salt &	Vinegar	175g
##			3103
##		Pringles Slt Vingar	134g
##			3095
##	Old El Paso Salsa	Dip Tomato Mild	300g
##			3085
##	Kettle Sensations	BBQ&Maple	150g
##			3083
##	Pringles Sthrn Fried	Chicken	134g
##			3083
##	Tostitos Lightly	Salted	175g
##			3074
##	Doritos Cheese	Supreme	330g
##			3052
##		Kettle Chilli	175g
##			3038
##	Smiths Chip Thinly	Cut Original	175g
##			1614
##	Snbts Whlgrn Crisps	Cheddr&Mstrd	90g
##			1576
##	Natural Chip Co	Tmato Hrb&Spce	175g
##			1572
##		Burger Rings	220g
##			1564
##	Natural ChipCo Sea	Salt & Vinegr	175g
##			1550
##		CCs Tasty Cheese	175g

##			1539
##	RRD SR Slow Rst	Pork Belly	150g
##			1526
##	Smiths Thinly Cut	Roast Chicken	175g
##			1519
##	RRD Sweet Chilli &	Sour Cream	165g
##			1516
##	Woolworths Cheese	Rings	190g
##			1516
##		CCs Original	175g
##			1514
##	RRD Honey Soy	Chicken	165g
##			1513
##	Smith Crinkle Cut	Mac N Cheese	150g
##			1512
##	WW Supreme Cheese	Corn Chips	200g
##			1509
##	Infuzions Mango	Chutny Papadums	70g
##			1507
##	RRD Chilli&	Coconut	150g
##			1506
##	Smiths Crinkle Cut	Snag&Sauce	150g
##			1503
##		CCs Nacho Cheese	175g
##			1498
##	Red Rock Deli Sp	Salt & Truffle	150G
##			1498
##	Red Rock Deli Thai	Chilli&Lime	150g
##			1495
##	WW Original Corn	Chips	200g
##			1495
##	Woolworths Mild	Salsa	300g
##			1491
##	Smiths Crinkle Cut	Chips Barbecue	170g
##			1489
##	WW Original Stacked Chips		160g
##			1487
##	Smiths Crinkle Cut	Chips Chicken	170g
##			1484
##	WW Sour Cream &Onion	Stacked Chips	160g
##			1483
##	Smiths Crinkle Cut	Chips Chs&Onion	170g
##			1481
##	Cheetos Chs & Bacon	Balls	190g
##			1479
##	RRD Salt & Vinegar		165g
##			1474
##	RRD Lime & Pepper		165g
##			1473
##	Smiths Chip Thinly	S/Cream&Onion	175g
##			1473
##	Doritos Salsa Mild		300g
##			1472
##	Smiths Crinkle Cut	Tomato Salsa	150g

```
##                                1470
##      WW D/Style Chip      Sea Salt 200g
##                                1469
##      GrnWves Plus Btroot & Chilli Jam 180g
##                                1468
##      Natural Chip          Compny SeaSalt175g
##                                1468
##      WW Crinkle Cut        Chicken 175g
##                                1467
##      Smiths Crinkle Cut    Chips Original 170g
##                                1461
##      Smiths Thinly         Swt Chli&S/Cream175G
##                                1461
##      Natural ChipCo        Hony Soy Chckn175g
##                                1460
##      Red Rock Deli SR      Salsa & Mzzrlla 150g
##                                1458
##      RRD Steak &          Chimuchurri 150g
##                                1455
##                                (Other)
##                                21550
```

```
productwords <- data.table(unlist(strsplit(unique(transactionData[,PROD_NAME])," ")))
setnames(productwords,"words")
clean_words <- productwords[!grepl("[0-9&/]",words),]
word_freq <- clean_words[, .N, by = words][order(-N)]
head(word_freq)
```

```
##      words      N
##      <char> <int>
## 1:      234
## 2: Chips      21
## 3: Smiths     16
## 4: Crinkle    14
## 5: Cut        14
## 6: Kettle     13
```

```
transactionData[, SALSA := grepl("salsa", tolower(PROD_NAME))]
transactionData <- transactionData[SALSA == FALSE,][, SALSA := NULL]
summary(transactionData)
```

```
##      DATE          STORE_NBR      LYLTY_CARD_NBR      TXN_ID
##  Min.   :2018-07-01  Min.    : 1.0  Min.    : 1000  Min.    :    1
## 1st Qu.:2018-09-30 1st Qu.: 70.0 1st Qu.: 70015 1st Qu.: 67569
## Median :2018-12-30 Median :130.0 Median : 130367 Median : 135183
## Mean   :2018-12-30 Mean   :135.1 Mean   : 135531 Mean   : 135131
## 3rd Qu.:2019-03-31 3rd Qu.:203.0 3rd Qu.: 203084 3rd Qu.: 202654
## Max.   :2019-06-30 Max.   :272.0 Max.   :2373711 Max.   :2415841
##      PROD_NBR      PROD_NAME      PROD_QTY      TOT_SALES
##  Min.    : 1.00  Length:246742  Min.    : 1.000  Min.    : 1.700
## 1st Qu.: 26.00  Class :character 1st Qu.: 2.000 1st Qu.: 5.800
## Median : 53.00  Mode  :character Median : 2.000 Median : 7.400
## Mean    : 56.35          Mean   : 1.908 Mean   : 7.321
```

```
## 3rd Qu.: 87.00          3rd Qu.: 2.000  3rd Qu.: 8.800
## Max.    :114.00        Max.    :200.000 Max.    :650.000
```

```
transactionData[PROD_QTY == 200,]
```

```
##      DATE STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR
##      <Date>      <int>          <int> <int>    <int>
## 1: 2018-08-19      226          226000 226201      4
## 2: 2019-05-20      226          226000 226210      4
##      PROD_NAME PROD_QTY TOT_SALES
##      <char>    <int>    <num>
## 1: Dorito Corn Chp Supreme 380g      200      650
## 2: Dorito Corn Chp Supreme 380g      200      650
```

```
filter(transactionData, LYLTY_CARD_NBR == 226000)
```

```
## Index: <PROD_QTY>
##      DATE STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR
##      <Date>      <int>          <int> <int>    <int>
## 1: 2018-08-19      226          226000 226201      4
## 2: 2019-05-20      226          226000 226210      4
##      PROD_NAME PROD_QTY TOT_SALES
##      <char>    <int>    <num>
## 1: Dorito Corn Chp Supreme 380g      200      650
## 2: Dorito Corn Chp Supreme 380g      200      650
```

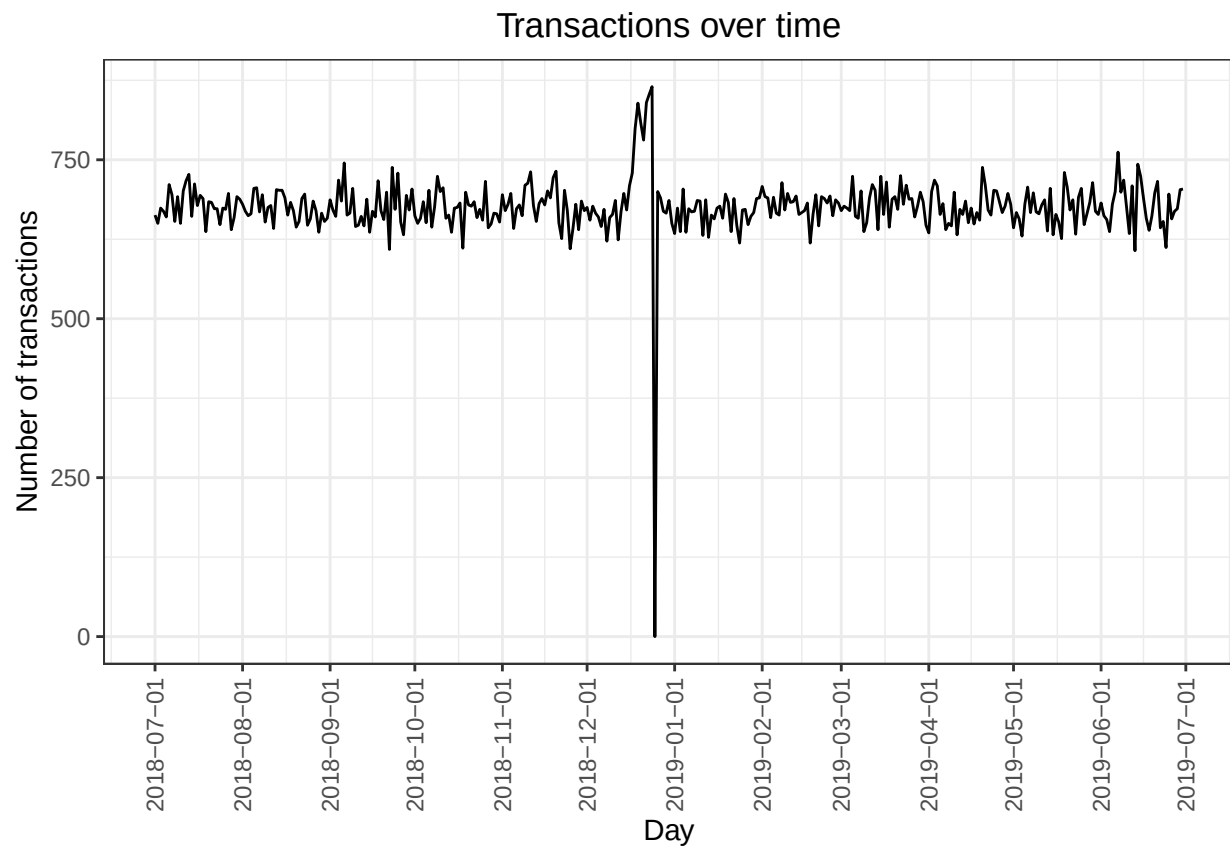
```
transactionData <- filter(transactionData, LYLTY_CARD_NBR != "226000")
summary(transactionData)
```

```
##      DATE          STORE_NBR      LYLTY_CARD_NBR      TXN_ID
## Min.   :2018-07-01  Min.    : 1.0  Min.     : 1000  Min.    :    1
## 1st Qu.:2018-09-30  1st Qu.: 70.0  1st Qu.: 70015  1st Qu.: 67569
## Median :2018-12-30  Median :130.0 Median : 130367 Median : 135182
## Mean   :2018-12-30  Mean   :135.1 Mean   : 135530 Mean   : 135130
## 3rd Qu.:2019-03-31  3rd Qu.:203.0 3rd Qu.: 203083 3rd Qu.: 202652
## Max.   :2019-06-30  Max.   :272.0 Max.   :2373711 Max.   :2415841
##      PROD_NBR      PROD_NAME      PROD_QTY      TOT_SALES
## Min.    : 1.00  Length:246740  Min.    :1.000  Min.    : 1.700
## 1st Qu.: 26.00  Class :character  1st Qu.:2.000  1st Qu.: 5.800
## Median : 53.00  Mode  :character  Median :2.000  Median : 7.400
## Mean    : 56.35              Mean   :1.906  Mean    : 7.316
## 3rd Qu.: 87.00              3rd Qu.:2.000  3rd Qu.: 8.800
## Max.    :114.00              Max.    :5.000  Max.    :29.500
```

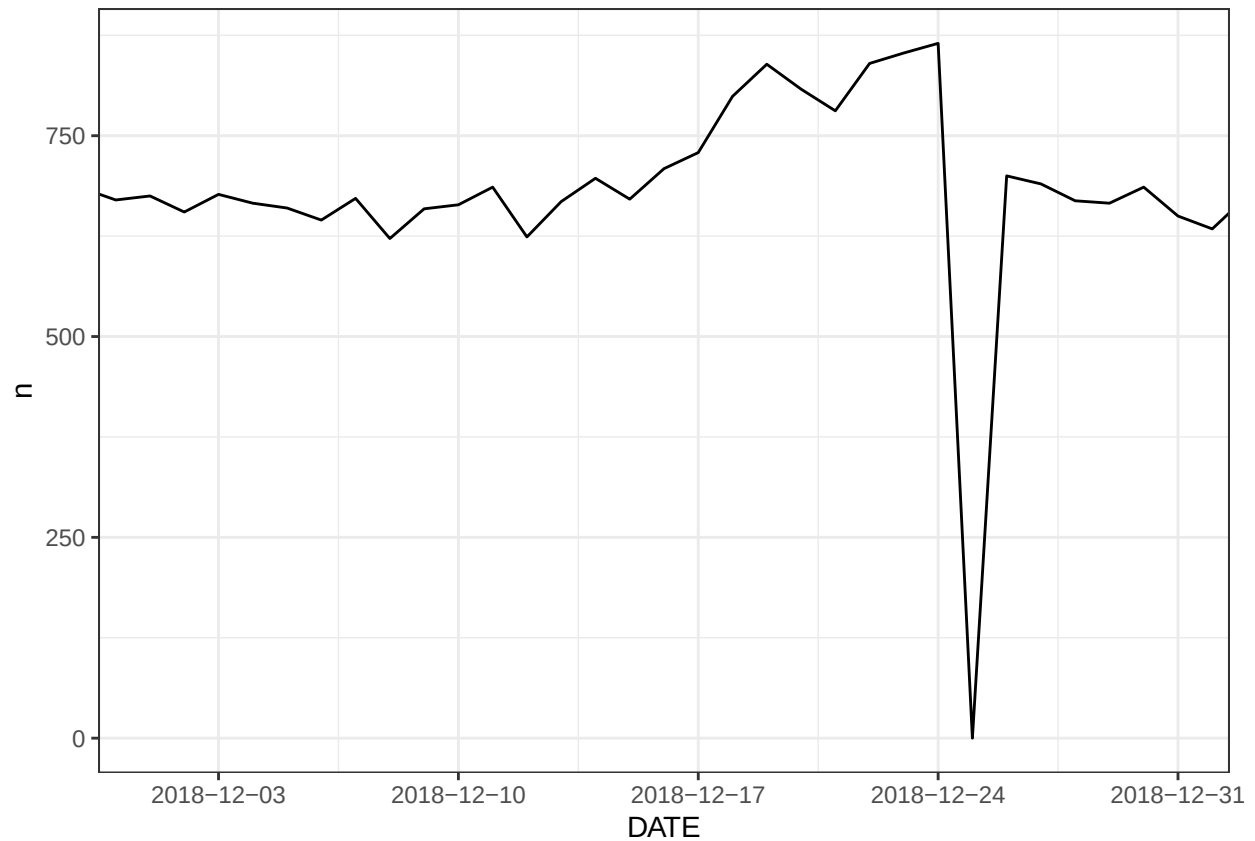
```
transaction_by_date <- table(transactionData$DATE)
head(transaction_by_date)
```

```
##
## 2018-07-01 2018-07-02 2018-07-03 2018-07-04 2018-07-05 2018-07-06
##      663      650      674      669      660      711
```

```
complete_data <- transactionData %>%
  count(DATE) %>%
  complete(DATE = seq(as.Date("2018-07-01"), as.Date("2019-06-30"), by = "day"), fill = list(n = 0))
theme_set(theme_bw())
theme_update(plot.title = element_text(hjust = 0.5))
ggplot(complete_data, aes(x = DATE, y = n)) +
  geom_line() +
  labs(x = "Day", y = "Number of transactions", title = "Transactions over time") +
  scale_x_date(breaks = "1 month") +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5))
```



```
ggplot(complete_data, aes(x = DATE, y = n)) +
  geom_line() +
  coord_cartesian(xlim = as.Date(c("2018-12-01", "2018-12-31"))) +
  scale_x_date(date_labels = "%Y-%m-%d")
```

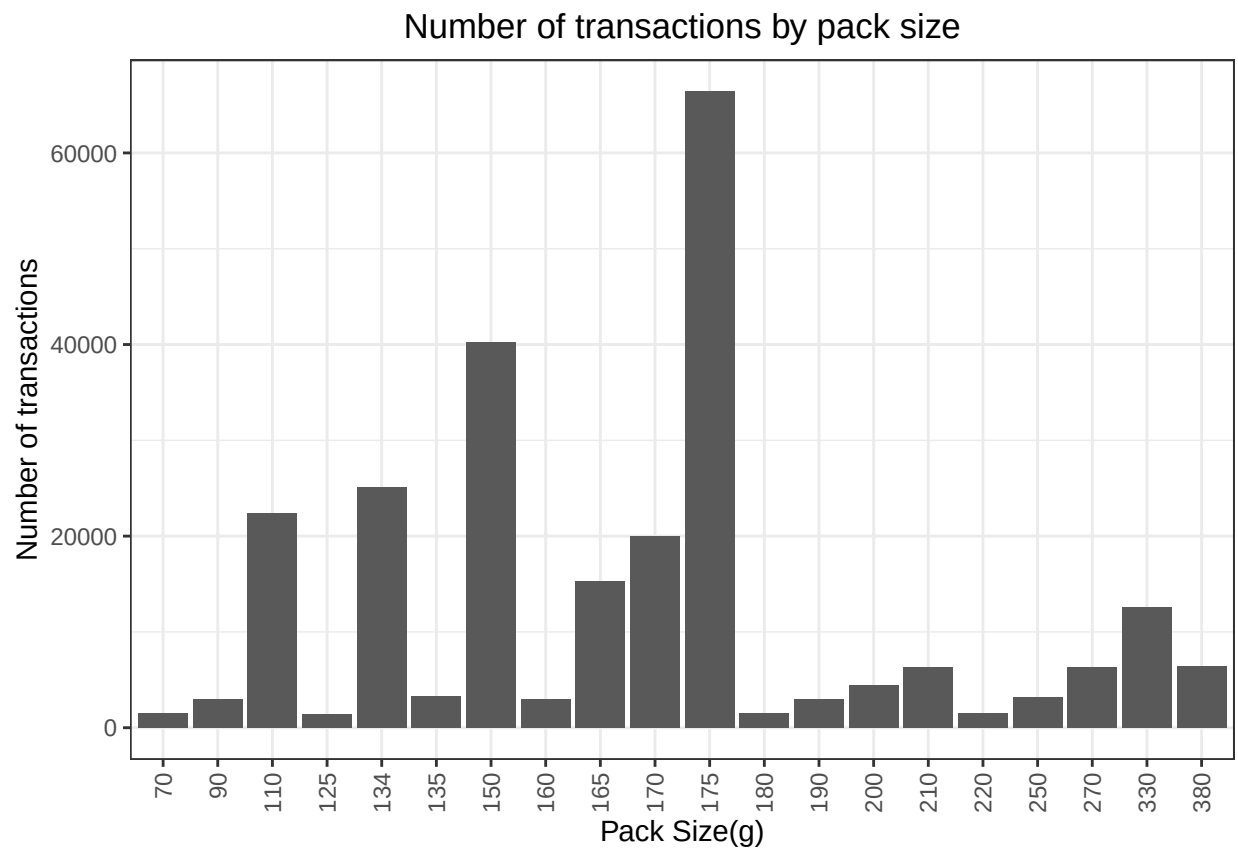


```
transactionData_table <- as.data.table(transactionData)
transactionData_table[, PACK_SIZE := parse_number(PROD_NAME)]
transactionData_table[, .N, PACK_SIZE][order(PACK_SIZE)]
```

```
##      PACK_SIZE      N
##      <num> <int>
## 1:         70  1507
## 2:         90  3008
## 3:        110 22387
## 4:        125  1454
## 5:        134 25102
## 6:        135  3257
## 7:        150 40203
## 8:        160  2970
## 9:        165 15297
## 10:       170 19983
## 11:       175 66390
## 12:       180  1468
## 13:       190  2995
## 14:       200  4473
## 15:       210  6272
## 16:       220  1564
## 17:       250  3169
## 18:       270  6285
## 19:       330 12540
```

```
## 20:      380  6416
##      PACK_SIZE      N
##      <num> <int>
```

```
ggplot(transactionData_table, aes(x = factor(PACK_SIZE))) +
  geom_bar() +
  labs(x = "Pack Size(g)", y = "Number of transactions", title = "Number of transactions by pack size")
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```



```
transactionData_table[, BRAND := word(PROD_NAME,1)]
transactionData_table[, .N, by = BRAND][order(-N)]
```

```
##      BRAND      N
##      <char> <int>
## 1:   Kettle 41288
## 2:   Smiths 27390
## 3: Pringles 25102
## 4:   Doritos 22041
## 5:    Thins 14075
## 6:    RRD   11894
## 7: Infuzions 11057
## 8:    WW    10320
## 9:    Cobs   9693
## 10: Tostitos  9471
## 11: Twisties  9454
```

```
## 12: Tyrrells 6442
## 13: Grain 6272
## 14: Natural 6050
## 15: Cheezels 4603
## 16: CCs 4551
## 17: Red 4427
## 18: Dorito 3183
## 19: Infzns 3144
## 20: Smith 2963
## 21: Cheetos 2927
## 22: Snbts 1576
## 23: Burger 1564
## 24: Woolworths 1516
## 25: GrnWves 1468
## 26: Sunbites 1432
## 27: NCC 1419
## 28: French 1418
## BRAND N
## <char> <int>
```

```
transactionData_table[BRAND == "Red", BRAND := "RRD"]
str(customerData)
```

```
## Classes 'data.table' and 'data.frame': 72637 obs. of 3 variables:
## $ LYLTY_CARD_NBR : int 1000 1002 1003 1004 1005 1007 1009 1010 1011 1012 ...
## $ LIFESTAGE : chr "YOUNG SINGLES/COUPLES" "YOUNG SINGLES/COUPLES" "YOUNG FAMILIES" "OLDER SI
## $ PREMIUM_CUSTOMER: chr "Premium" "Mainstream" "Budget" "Mainstream" ...
## - attr(*, ".internal.selfref")=<externalptr>
```

```
summary(customerData)
```

```
## LYLTY_CARD_NBR LIFESTAGE PREMIUM_CUSTOMER
## Min. : 1000 Length:72637 Length:72637
## 1st Qu.: 66202 Class :character Class :character
## Median : 134040 Mode :character Mode :character
## Mean : 136186
## 3rd Qu.: 203375
## Max. : 2373711
```

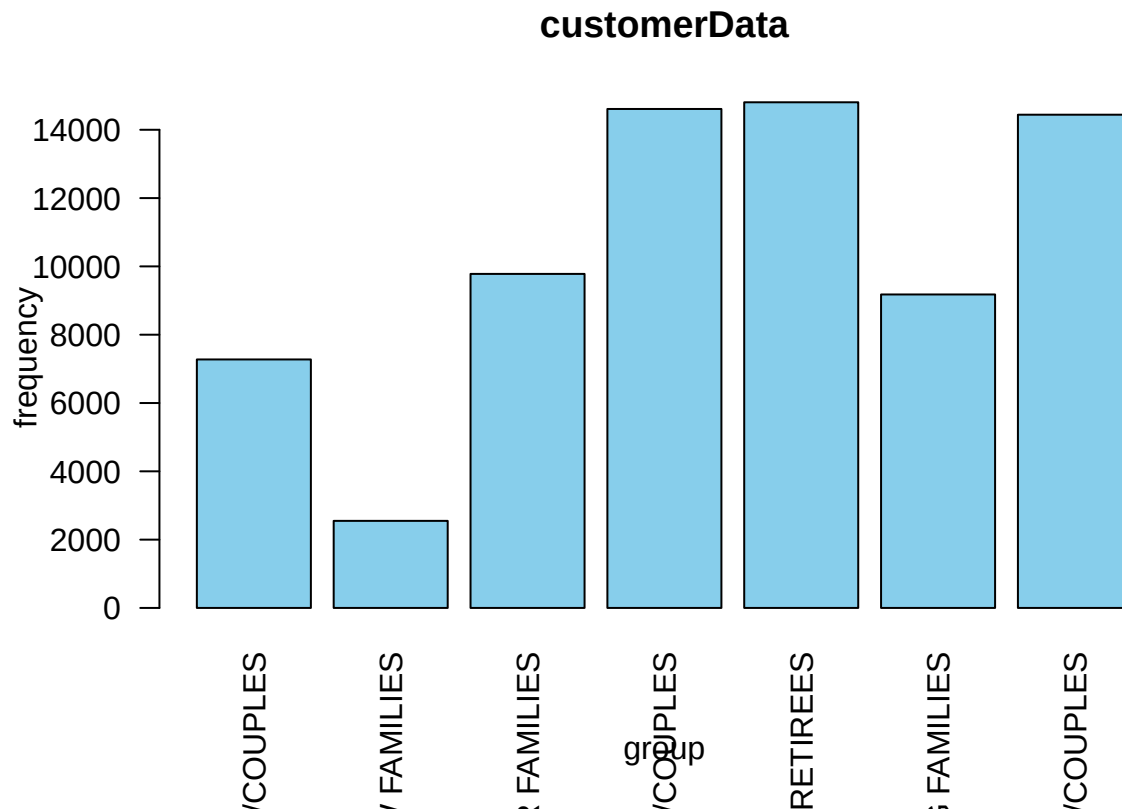
```
head(customerData)
```

```
## LYLTY_CARD_NBR LIFESTAGE PREMIUM_CUSTOMER
## <int> <char> <char>
## 1: 1000 YOUNG SINGLES/COUPLES Premium
## 2: 1002 YOUNG SINGLES/COUPLES Mainstream
## 3: 1003 YOUNG FAMILIES Budget
## 4: 1004 OLDER SINGLES/COUPLES Mainstream
## 5: 1005 MIDAGE SINGLES/COUPLES Mainstream
## 6: 1007 YOUNG SINGLES/COUPLES Budget
```

```
table(customerData$PREMIUM_CUSTOMER)
```

```
##
##      Budget Mainstream      Premium
##      24470      29245      18922
```

```
customerData_lifestage_table <-table(customerData$LIFESTAGE)
barplot(customerData_lifestage_table,
        main = "customerData",
        xlab = "group",
        ylab = "frequency",
        col = "skyblue",
        las = 2)
```



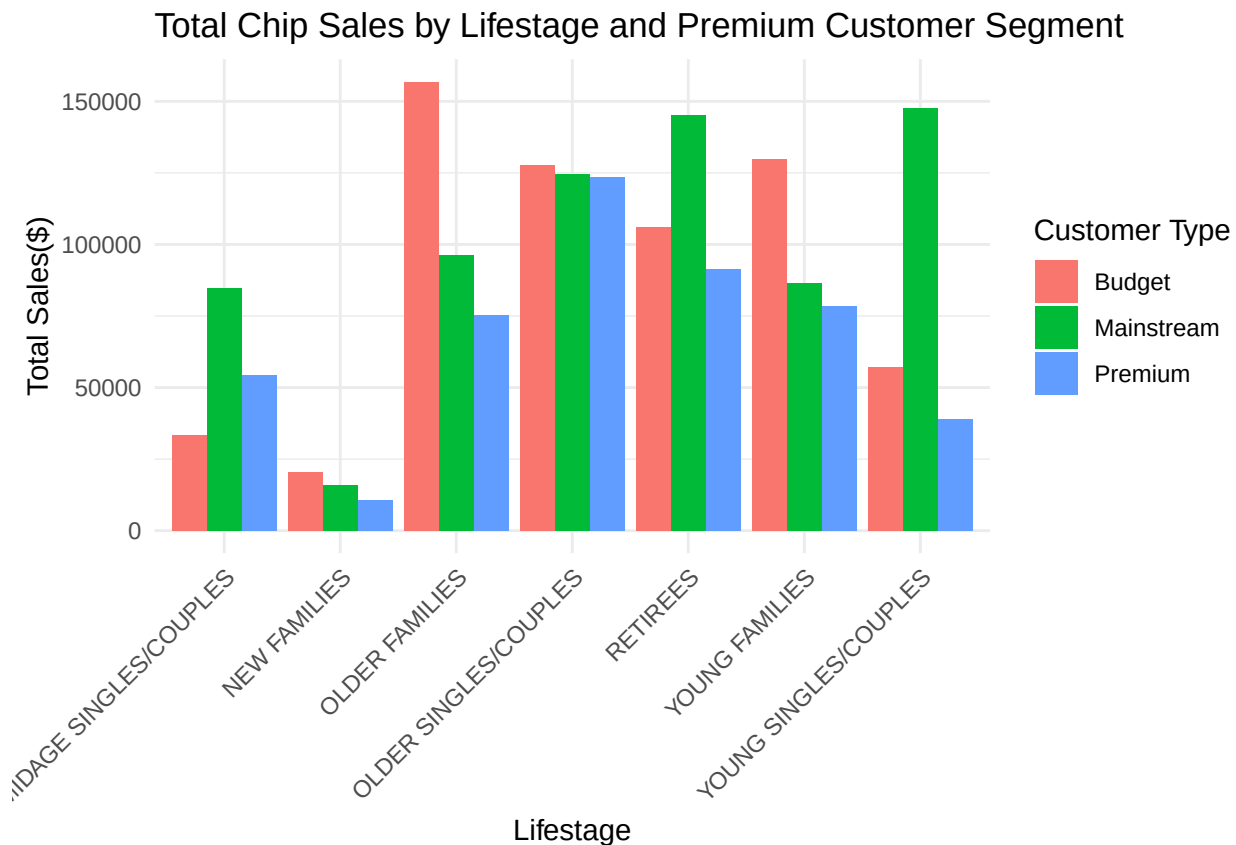
```
data <- merge(transactionData,customerData, all.x = TRUE)
sum(is.na(data))
```

```
## [1] 0
```

```
fwrite(data,paste0(filePath, "QVI_data.csv"))
sale_summary <- data %>%
  group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
  summarise(Total_sales = sum(TOT_SALES, na.rm = TRUE), .group = "drop")
```

```
## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by LIFESTAGE and PREMIUM_CUSTOMER.
## i Output is grouped by LIFESTAGE.
## i Use 'summarise(groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(LIFESTAGE, PREMIUM_CUSTOMER))' for per-operation
## grouping ('?dplyr::dplyr_by') instead.
```

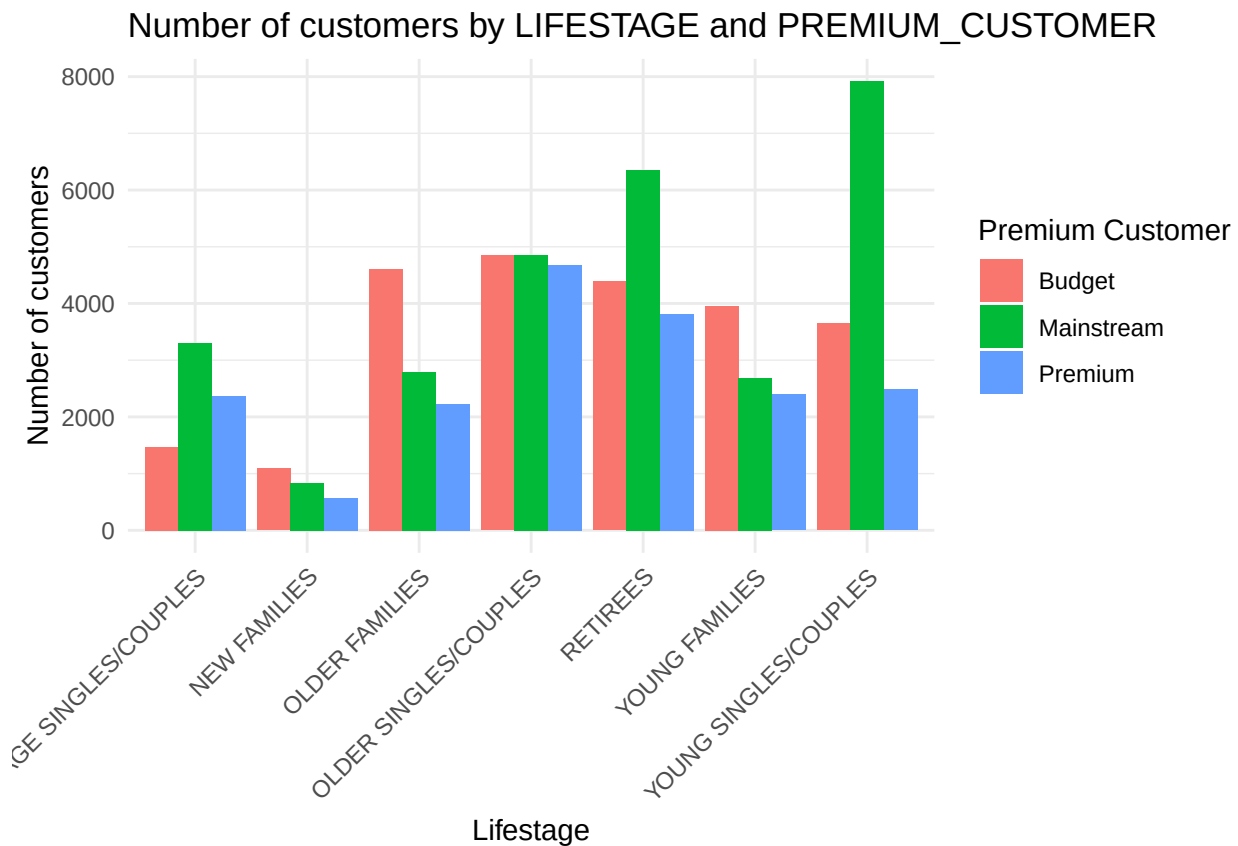
```
ggplot(sale_summary, aes(x = LIFESTAGE, y = Total_sales, fill = PREMIUM_CUSTOMER)) +
  geom_col(position = "dodge") +
  labs(title = "Total Chip Sales by Lifestage and Premium Customer Segment",
       x = "Lifestage",
       y = "Total Sales($)",
       fill = "Customer Type") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
customer_summary <- data %>%
  group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
  summarise(customer_count = n_distinct(LYLT_CARD_NBR), .group = "drop")
```

```
## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by LIFESTAGE and PREMIUM_CUSTOMER.
## i Output is grouped by LIFESTAGE.
## i Use 'summarise(groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(LIFESTAGE, PREMIUM_CUSTOMER))' for per-operation
## grouping ('?dplyr::dplyr_by') instead.
```

```
ggplot(customer_summary, aes(x = LIFESTAGE, y = customer_count, fill = PREMIUM_CUSTOMER)) +
  geom_col(position = "dodge") +
  labs(title = "Number of customers by LIFESTAGE and PREMIUM_CUSTOMER",
       x = "Lifestage",
       y = "Number of customers",
       fill = "Premium Customer") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



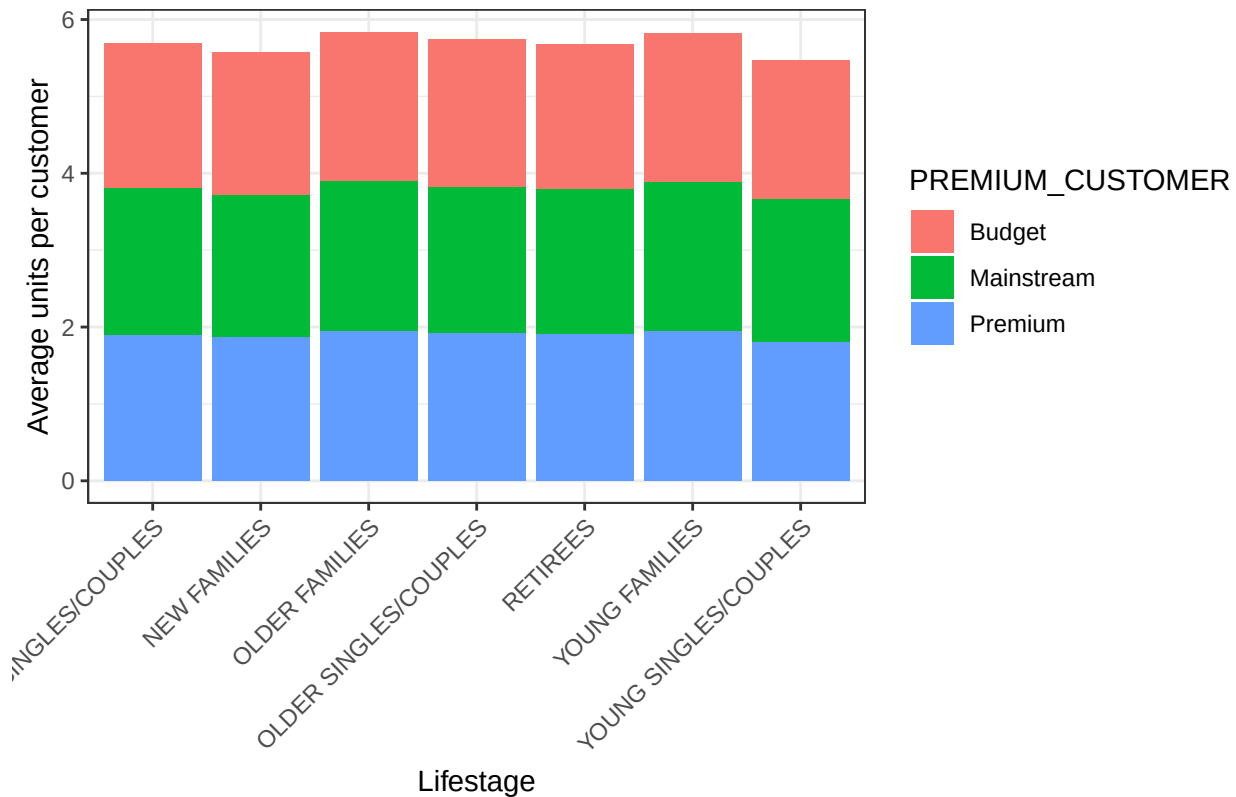
```
avg_units <- data %>%
  group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
  summarise(avg_units_per_customer = mean(PROD_QTY), .group = "drop")
```

```
## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by LIFESTAGE and PREMIUM_CUSTOMER.
## i Output is grouped by LIFESTAGE.
## i Use 'summarise(.groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(LIFESTAGE, PREMIUM_CUSTOMER))' for per-operation
## grouping ('?dplyr::dplyr_by') instead.
```

```
ggplot(avg_units, aes(x = LIFESTAGE, y = avg_units_per_customer, fill = PREMIUM_CUSTOMER)) +
  geom_col() +
  labs(title = "Average number of units of per customer by LIFESTAGE and PREMIUM_CUSTOMER",
       y = "Average units per customer",
```

```
x = "Lifestage") +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

iber of units of per customer by LIFESTAGE and PREMIUM_CUSTOMER

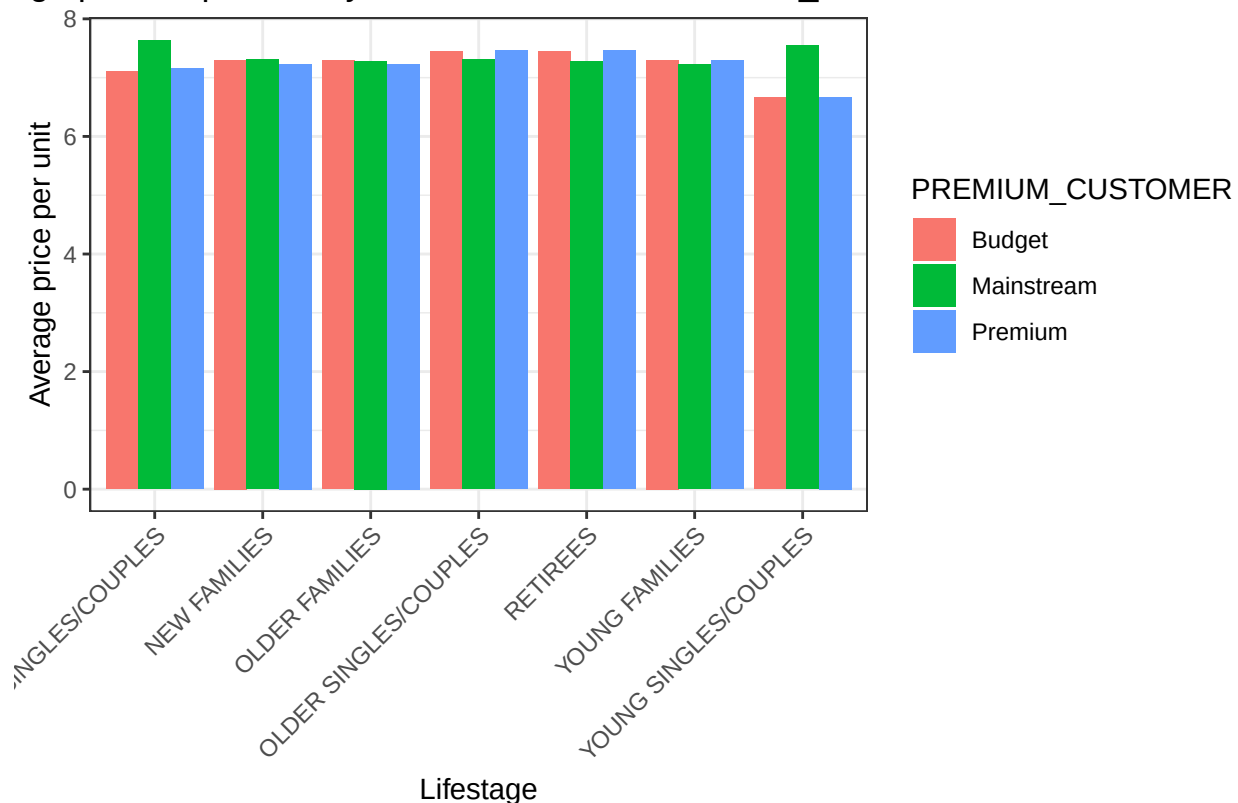


```
avg_price <- data %>%
  group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
  summarise(avg_price_per_unit = mean(TOT_SALES), .group = "drop")
```

```
## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by LIFESTAGE and PREMIUM_CUSTOMER.
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## i Use 'summarise(.by = c(LIFESTAGE, PREMIUM_CUSTOMER))' for per-operation
## grouping ('?dplyr::dplyr_by') instead.
```

```
ggplot(avg_price, aes(x = LIFESTAGE, y = avg_price_per_unit, fill = PREMIUM_CUSTOMER)) +
  geom_col(position = "dodge") +
  labs(title = "Average price of per unit by LIFESTAGE and PREMIUM_CUSTOMER",
       y = "Average price per unit",
       x = "Lifestage") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

age price of per unit by LIFESTAGE and PREMIUM_CUSTOMER



```
groupA <- subset(avg_price, (PREMIUM_CUSTOMER == "Mainstream" & LIFESTAGE == "MIDAGE SINGLES/COUPLES"))
groupB <- subset(avg_price, (PREMIUM_CUSTOMER == "Budget" & LIFESTAGE == "MIDAGE SINGLES/COUPLES")) | (PREMIUM_CUSTOMER == "Premium" & LIFESTAGE == "MIDAGE SINGLES/COUPLES"))
groupC <- subset(avg_price, (PREMIUM_CUSTOMER == "Premium" & LIFESTAGE == "MIDAGE SINGLES/COUPLES")) | (PREMIUM_CUSTOMER == "Budget" & LIFESTAGE == "MIDAGE SINGLES/COUPLES"))
t.test(groupA$avg_price_per_unit, groupB$avg_price_per_unit)
```

```
##
## Welch Two Sample t-test
##
## data: groupA$avg_price_per_unit and groupB$avg_price_per_unit
## t = 3.1237, df = 1.0742, p-value = 0.1832
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -1.742855 3.159825
## sample estimates:
## mean of x mean of y
## 7.594218 6.885733
```

```
t.test(groupA$avg_price_per_unit, groupC$avg_price_per_unit)
```

```
##
## Welch Two Sample t-test
##
## data: groupA$avg_price_per_unit and groupC$avg_price_per_unit
## t = 2.8001, df = 1.0642, p-value = 0.2061
```

```
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -2.002782  3.365521
## sample estimates:
## mean of x mean of y
## 7.594218  6.912848
```

```
data1 <- merge(transactionData_table, customerData, all.x = TRUE)
target_customers <- subset(data1, PREMIUM_CUSTOMER == "Mainstream" & LIFESTAGE == "YOUNG SINGLES/COUPLES")
brand_preference <- table(target_customers$BRAND)
brand_preference_sorted <- sort(brand_preference, decreasing = TRUE)
size_preference <- table(target_customers$PACK_SIZE)
size_preference_sorted <- sort(size_preference, decreasing = TRUE)
```

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.